CHAPTER 20 · DERIVATIONS

OSCILLATIONS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 VELOCITY IN TERMS OF DISPLACEMENT

SYLLABUS 20.1(C)

RESULT $v = \pm\omega\sqrt{A^2 - x^2}$

1. **Method 1 (from $a = -\omega^2 x$)**: write $a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$, so $v\frac{dv}{dx} = -\omega^2 x$.
2. Separate and integrate: $\int v\,dv = -\omega^2 \int x\,dx$, using $v = 0$ when $x = A$ (extreme position).
3. $\frac{1}{2}v^2 = \frac{1}{2}\omega^2(A^2 - x^2)$, so $\mathbf{v = \pm\omega\sqrt{A^2 - x^2}}$.
4. **Method 2 (from $x = A\sin\omega t$)**: $\sin\omega t = \frac{x}{A}$ and $v = \frac{dx}{dt} = A\omega\cos\omega t$, so $\cos\omega t = \frac{v}{A\omega}$.
5. Use $\sin^2\omega t + \cos^2\omega t = 1$: $\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1$, which again gives $\mathbf{v = \pm\omega\sqrt{A^2 - x^2}}$; maximum ωA at $x = 0$, zero at $x = \pm A$.

2 ACCELERATION IN TERMS OF TIME, AND SHOWING THE MOTION IS SHM

SYLLABUS 20.2

RESULT $a = -\omega^2 A\sin\omega t = -\omega^2 x$

1. Given the displacement $x = A\sin\omega t$.
2. Differentiate x with respect to t: $v = \frac{dx}{dt} = A\omega\cos\omega t$.
3. Differentiate v with respect to t: $\mathbf{a = \frac{dv}{dt} = -A\omega^2\sin\omega t}$.
4. Since $A\sin\omega t = x$, $\mathbf{a = -\omega^2 x}$: the acceleration is directly proportional to the displacement and always directed towards the equilibrium position, so the motion is SHM.

3 KINETIC AND POTENTIAL ENERGY IN SHM

SYLLABUS 20.2(A)

RESULT $K = \frac{1}{2}m\omega^2(A^2 - x^2)$, $U = \frac{1}{2}m\omega^2 x^2$

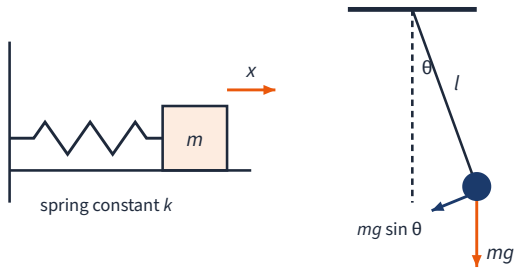
1. Use $K = \frac{1}{2}mv^2$ with $v^2 = \omega^2(A^2 - x^2)$: $\mathbf{K = \frac{1}{2}m\omega^2(A^2 - x^2)}$.
2. K is maximum at $x = 0$, where $U = 0$, so the total energy is $E = K_{max} = \frac{1}{2}m\omega^2 A^2$.
3. Principle of conservation of energy at any position: $U = E - K = \frac{1}{2}m\omega^2 A^2 - \frac{1}{2}m\omega^2(A^2 - x^2) = \mathbf{\frac{1}{2}m\omega^2 x^2}$.
4. In terms of t ($x = A\sin\omega t$): $K = \frac{1}{2}m\omega^2 A^2\cos^2\omega t$ and $U = \frac{1}{2}m\omega^2 A^2\sin^2\omega t$; $K + U = \frac{1}{2}m\omega^2 A^2$ is constant.

4 SHOWING A SYSTEM PERFORMS SHM: SPRING-MASS SYSTEM AND SIMPLE PENDULUM

SYLLABUS 20.3(A)

RESULT $T = 2\pi\sqrt{\frac{m}{k}}$, $T = 2\pi\sqrt{\frac{l}{g}}$

1. **Method**: write the restoring (or resultant) force on the body at displacement x, put it equal to ma, and show that a = −(constant)x. Then ω^2 = constant and $T = \frac{2\pi}{\omega}$.
2. **Spring–mass**: displaced x from equilibrium, restoring force F = −kx. ma = −kx, so $a = -\frac{k}{m}x$.
3. This is SHM ($a = -\omega^2 x$) with $\omega^2 = \frac{k}{m}$: $\mathbf{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}}$.
4. **Simple pendulum**: displaced by angle θ, restoring force along the arc = −mg sin θ ≈ −mgθ for small θ, and $\theta = \frac{x}{l}$.
5. $ma = -\frac{mgx}{l}$, so $a = -\frac{g}{l}x$: $\omega^2 = \frac{g}{l}$ and $\mathbf{T = 2\pi\sqrt{\frac{l}{g}}}$.



Valid for small oscillations (θ less than about 10°).

CHAPTER 21 · DERIVATIONS

WAVE MOTION

3
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 PHASE DIFFERENCE

SYLLABUS 21.1(D)

RESULT $\varphi = kx = \dfrac{2\pi x}{\lambda}$

1. Two points one wavelength apart differ in phase by 2π.
2. Phase difference is proportional to separation: $\dfrac{\varphi}{2\pi} = \dfrac{x}{\lambda}$.
3. $\boldsymbol{\varphi = \dfrac{2\pi x}{\lambda} = kx}$, with $k = \dfrac{2\pi}{\lambda}$.

2 WAVE SPEED

SYLLABUS 21.1(E)

RESULT $v = f\lambda$

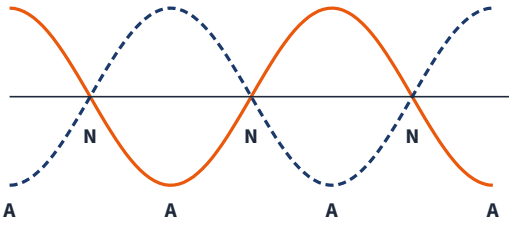
1. In one period T the wave moves forward one wavelength λ.
2. $v = \dfrac{\lambda}{T}$, and $f = \dfrac{1}{T}$, so $\boldsymbol{v = f\lambda}$ (equivalently $v = \dfrac{\omega}{k}$).

3 STANDING WAVE EQUATION

SYLLABUS 21.4(B)

RESULT $y = 2A \cos kx \sin \omega t$

1. Two identical waves travel in opposite directions: $y_1 = A \sin(\omega t - kx)$, $y_2 = A \sin(\omega t + kx)$.
2. Superposition: $y = y_1 + y_2$. Use $\sin P + \sin Q = 2 \sin \dfrac{P+Q}{2} \cos \dfrac{P-Q}{2}$.
3. $\boldsymbol{y = 2A \cos kx \sin \omega t}$.
4. Amplitude $2A \cos kx$ depends on position: nodes where $\cos kx = 0$ ($x = \dfrac{\lambda}{4}, \dfrac{3\lambda}{4}, \ldots$), antinodes where $|\cos kx| = 1$ ($x = 0, \dfrac{\lambda}{2}, \lambda, \ldots$). Adjacent nodes are $\dfrac{\lambda}{2}$ apart; no energy is carried along.



N = node, A = antinode; N to N = λ/2

CHAPTER 22 · DERIVATIONS

SOUND WAVES

3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FUNDAMENTAL FREQUENCY: STRETCHED STRING, CLOSED PIPE AND OPEN PIPE

SYLLABUS 22.2

RESULT $f = \frac{v}{2L}$, $f = \frac{v}{4L}$, $f = \frac{v}{2L}$

1. Draw the fundamental standing wave for each system, then use length of the system = number of wavelengths in it, and $f = \frac{v}{\lambda}$.
2. **Stretched string** (length L, node at each fixed end, one antinode in the middle): $L = \frac{\lambda}{2}$, so $\lambda = 2L$ and $\mathbf{f = \frac{v}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}}}$, with $v = \sqrt{\frac{T}{\mu}}$.
3. **Closed pipe** (node at the closed end, antinode at the open end): $L = \frac{\lambda}{4}$, so $\lambda = 4L$ and $\mathbf{f = \frac{v}{4L}}$.
4. **Open pipe** (antinode at each end, one node in the middle): $L = \frac{\lambda}{2}$, so $\lambda = 2L$ and $\mathbf{f = \frac{v}{2L}}$.

> End correction is ignored. Overtones: string and open pipe give all harmonics ($f_n = \frac{nv}{2L}$); a closed pipe gives odd harmonics only ($f_n = \frac{(2n-1)v}{4L}$).

2 BEAT FREQUENCY

SYLLABUS 22.4(B)

RESULT $f = f_1 - f_2$

1. At a point: $y_1 = A \sin 2\pi f_1 t$ and $y_2 = A \sin 2\pi f_2 t$.
2. $y = y_1 + y_2 = 2A \cos(2\pi \frac{f_1 - f_2}{2} t) \sin(2\pi \frac{f_1 + f_2}{2} t)$.
3. The resultant sound has the average frequency $\frac{f_1 + f_2}{2}$ and a slowly varying amplitude $A_R = 2A \cos(\pi(f_1 - f_2)t)$.
4. The amplitude is maximum ($|A_R| = 2A$, loud sound) when $\cos(\pi(f_1 - f_2)t) = \pm 1$, i.e. $\pi(f_1 - f_2)t = 0, \pi, 2\pi, \ldots$ so $t = 0, \frac{1}{f_1 - f_2}, \frac{2}{f_1 - f_2}, \ldots$
5. The time between two successive loud sounds (period of the beat) is $T = \frac{1}{f_1 - f_2}$.
6. Beat frequency = $\frac{1}{T}$: $\mathbf{f = f_1 - f_2}$.

> The syllabus notes that this derivation is required.

3 DOPPLER EFFECT (FOUR CASES)

SYLLABUS 22.5(B)

RESULT $f' = \dfrac{f(v \pm u_o)}{v \mp u_s}$

1. Stationary source of frequency f; speed of sound v; wavelength $\lambda = \dfrac{v}{f}$.

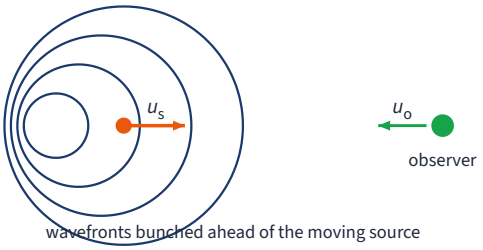
2. **(a) Observer moves towards the source** at u_o: the wavelength is unchanged, but the sound passes the observer at relative speed $v + u_o$, so $f' = \dfrac{v + u_o}{\lambda} = \mathbf{\dfrac{f(v + u_o)}{v}}$.

3. **(b) Observer moves away from the source**: relative speed $v - u_o$, so $\mathbf{f' = \dfrac{f(v - u_o)}{v}}$.

4. **(c) Source moves towards the observer** at u_s: in one period $\dfrac{1}{f}$ the wave moves $\dfrac{v}{f}$ while the source moves $\dfrac{u_s}{f}$, so the wavelength in front is shortened to $\lambda' = \dfrac{v - u_s}{f}$. The speed of sound is unchanged: $f' = \dfrac{v}{\lambda'} = \mathbf{\dfrac{fv}{v - u_s}}$.

5. **(d) Source moves away from the observer**: $\lambda' = \dfrac{v + u_s}{f}$, so $\mathbf{f' = \dfrac{fv}{v + u_s}}$.

6. Both moving: combine the two effects, $\mathbf{f' = \dfrac{f(v \pm u_o)}{v \mp u_s}}$ (upper signs when approaching).



wavefronts bunched ahead of the moving source

Observer moving: the wavelength stays the same, the relative speed changes. Source moving: the wavelength changes, the speed of sound stays the same.

CHAPTER 23 · DERIVATIONS

GEOMETRICAL OPTICS

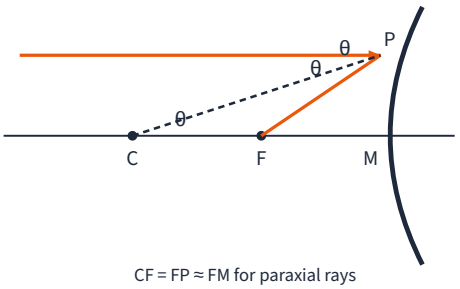
6 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FOCAL LENGTH OF A SPHERICAL MIRROR

SYLLABUS 23.1(A)

RESULT $f = \frac{r}{2}$

1. A ray parallel to the axis strikes a concave mirror at P. The normal at P passes through the centre of curvature C.
2. Angle of incidence = angle of reflection = θ. The parallel ray makes angle θ with CP too (alternate angles), so triangle CFP has two equal angles.
3. CF = FP (isosceles triangle).
4. For paraxial rays P is close to the pole M, so FP ≈ FM. Then CM = CF + FM ≈ 2FM.
5. r = 2f, so $\mathbf{f = \frac{r}{2}}$.



CF = FP ≈ FM for paraxial rays

Use the sign convention consistently.

2 MIRROR EQUATION (REFLECTION AT A CURVED SURFACE)

SYLLABUS 23.1

RESULT $\frac{1}{u} + \frac{1}{v} = \frac{2}{r} = \frac{1}{f}$

1. A ray from a point object O on the axis strikes a concave mirror at P (height h above the axis) and reflects to the image I. The normal at P passes through the centre of curvature C.
2. Let the angles that OP, CP and IP make with the axis be α, β and γ. The angle of incidence is θ = β − α and the angle of reflection is θ = γ − β.
3. So β − α = γ − β, which gives α + γ = 2β.
4. Rays are very close to the principal axis (paraxial), so the angles are small: $\alpha \approx \frac{h}{u}, \beta \approx \frac{h}{r}, \gamma \approx \frac{h}{v}$.
5. $\frac{h}{u} + \frac{h}{v} = \frac{2h}{r}$, so $\mathbf{\frac{1}{u} + \frac{1}{v} = \frac{2}{r} = \frac{1}{f}}$.

Alternative (geometrical) method: use two pairs of similar triangles and relate their sides to the height h.

3 EQUATION EQUIVALENT TO SNELL'S LAW FOR PLANE SURFACES

SYLLABUS 23.2

RESULT $n_1 \sin\theta_1 = n_2 \sin\theta_2$

1. A ray in air at angle θ_a passes through two parallel-sided layers of refractive indices n_1 and n_2 and emerges into air again.
2. The surfaces are parallel, so the ray leaves into air at the same angle θ_a as it entered.
3. Air → medium 1: $n_1 = \dfrac{\sin\theta_a}{\sin\theta_1}$, so $\sin\theta_a = n_1 \sin\theta_1$.
4. Medium 2 → air: $n_2 = \dfrac{\sin\theta_a}{\sin\theta_2}$, so $\sin\theta_a = n_2 \sin\theta_2$.
5. Relate the angles: $\mathbf{n_1 \sin\theta_1 = n_2 \sin\theta_2}$ (n sin θ is constant across plane boundaries).

4 REFRACTION AT A CURVED SURFACE

SYLLABUS 23.2

RESULT $\dfrac{n_1}{u} + \dfrac{n_2}{v} = \dfrac{n_2 - n_1}{r}$

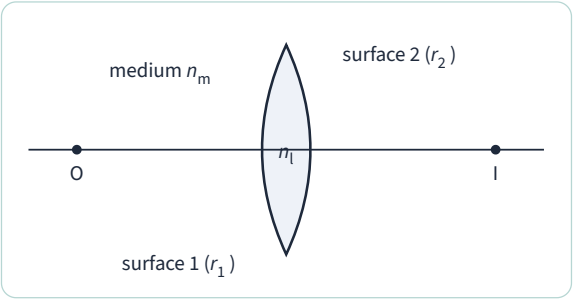
1. A ray from a point object O in medium n_1 strikes a convex spherical surface at P (height h) and refracts into medium n_2 to the image I. The normal at P passes through the centre of curvature C.
2. Let the angles that OP, CP and IP make with the axis be α, β and γ. Angle of incidence $\theta_1 = \alpha + \beta$; angle of refraction $\theta_2 = \beta - \gamma$.
3. Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$. For paraxial rays the angles are small: $n_1\theta_1 = n_2\theta_2$.
4. $n_1(\alpha + \beta) = n_2(\beta - \gamma)$, so $n_1\alpha + n_2\gamma = (n_2 - n_1)\beta$.
5. Small angles: $\alpha \approx \dfrac{h}{u}$, $\beta \approx \dfrac{h}{r}$, $\gamma \approx \dfrac{h}{v}$, so $\mathbf{\dfrac{n_1}{u} + \dfrac{n_2}{v} = \dfrac{n_2 - n_1}{r}}$.

5 THIN LENS FORMULA AND LENSMAKER'S EQUATION

SYLLABUS 23.3(A)

RESULT $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$; $\frac{1}{f_m}=(\frac{n_l}{n_m}-1)(\frac{1}{r_1}-\frac{1}{r_2})$

1. Lens of refractive index n_l in a medium n_m. Use $\frac{n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{r}$ at each surface.
2. Surface 1 (medium → lens): $\frac{n_m}{u}+\frac{n_l}{v_1}=\frac{n_l-n_m}{r_1}$ … (1)
3. The image I_1 is the object for surface 2. The lens is thin, so it is a virtual object at $u_2=-v_1$.
4. Surface 2 (lens → medium): $-\frac{n_l}{v_1}+\frac{n_m}{v}=\frac{n_m-n_l}{r_2}$ … (2)
5. Add (1) and (2): $\frac{n_m}{u}+\frac{n_m}{v}=(n_l-n_m)(\frac{1}{r_1}-\frac{1}{r_2})$.
6. Divide by n_m: $\frac{1}{u}+\frac{1}{v}=(\frac{n_l}{n_m}-1)(\frac{1}{r_1}-\frac{1}{r_2})$.
7. Object at infinity: $v=f_m$, so $\mathbf{\frac{1}{f_m}=(\frac{n_l}{n_m}-1)(\frac{1}{r_1}-\frac{1}{r_2})}$ (lensmaker's equation), and therefore $\mathbf{\frac{1}{u}+\frac{1}{v}=\frac{1}{f}}$.



In this form r is positive when the centre of curvature is on the side the light goes to, so for a biconvex lens r_2 is negative. Many books write $(\frac{1}{r_1}+\frac{1}{r_2})$ with both radii of a convex lens positive; both give the same f.

6 MINIMUM DISTANCE BETWEEN A REAL OBJECT AND ITS REAL IMAGE IS 4F

SYLLABUS 23.3

RESULT $d_{min}=4f$

1. Let the distance between the real object and the real image be d, so $u+v=d$ and $v=d-u$.
2. Substitute into the thin lens equation: $\frac{1}{u}+\frac{1}{d-u}=\frac{1}{f}$, so $d/(u(d-u))=\frac{1}{f}$.
3. Rearrange as a quadratic in u: $u^2-du+df=0$.
4. For a real image, u must be real: $b^2-4ac\geq 0$, so $d^2-4df\geq 0$ and $d\geq 4f$.
5. $\mathbf{d_{min}=4f}$, when $u=v=2f$.

CHAPTER 24 · DERIVATIONS

WAVE OPTICS

7 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 OPTICAL PATH LENGTH

SYLLABUS 24.2

RESULT optical path = nd

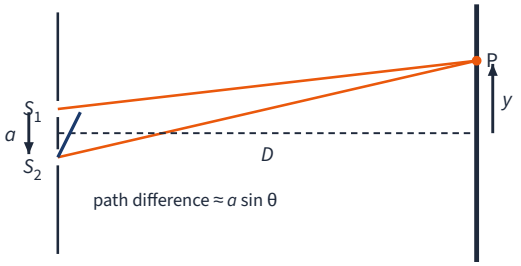
1. Light travels a distance d in a medium of refractive index n at speed $v = \frac{c}{n}$.
2. Time taken in the medium: $t = \frac{d}{v} = \frac{nd}{c}$.
3. In the same time t, light in vacuum travels $ct = c\frac{nd}{c}$.
4. **Optical path = nd**: the equivalent distance in vacuum. Optical path difference = $n_2d_2 - n_1d_1$.

2 FRINGE SEPARATION IN YOUNG'S EXPERIMENT

SYLLABUS 24.3(B)

RESULT $x = \frac{\lambda D}{a}$

1. Slits S_1 and S_2 a apart; screen at distance D ($D \gg a$). Point P is at height y above the centre.
2. Path difference $S_2P - S_1P \approx a \sin\theta \approx a \tan\theta = \frac{ay}{D}$ (small θ).
3. Bright fringe: $\frac{ay}{D} = m\lambda$, so $y_m = \frac{m\lambda D}{a}$.
4. Fringe separation $\mathbf{x = y_{m+1} - y_m = \frac{\lambda D}{a}}$.



3 THIN FILM INTERFERENCE

SYLLABUS 24.4

RESULT $2nt = (m + \frac{1}{2})\lambda$ bright, $2nt = m\lambda$ dark

1. Light falls normally on a thin film of thickness t and refractive index n in air. Ray 1 reflects at the top surface; ray 2 reflects at the bottom surface.
2. Optical path difference between the two reflected rays = 2nt.
3. Phase change: ray 1 reflects from a denser medium (air → film), so it has a phase change of π (equivalent to $\frac{\lambda}{2}$); ray 2 reflects from a less dense medium (film → air), so no phase change.
4. Constructive (bright): $2nt + \frac{\lambda}{2} = (m + 1)\lambda$, so $\mathbf{2nt = (m + \frac{1}{2})\lambda}$.
5. Destructive (dark): $2nt + \frac{\lambda}{2} = (m + \frac{1}{2})\lambda$, so $\mathbf{2nt = m\lambda}$ ($m = 0, 1, 2, \ldots$).

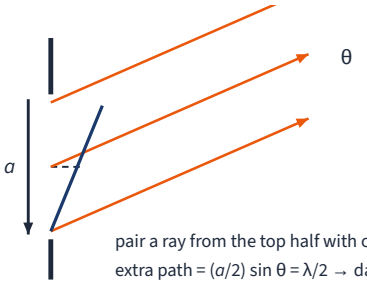
If both reflections have a phase change (e.g. a coating with n between air and glass), the two conditions swap: $2nt = m\lambda$ bright, $2nt = (m + \frac{1}{2})\lambda$ dark.

4 FIRST MINIMUM OF A SINGLE SLIT

SYLLABUS 24.5(B)

RESULT $\sin\theta = \frac{\lambda}{a}$

1. Divide the slit of width a into two halves.
2. Every point in the top half pairs with a point $\frac{a}{2}$ below it. In direction θ the path difference of each pair is $\frac{a}{2}\sin\theta$.
3. If $\frac{a}{2}\sin\theta = \frac{\lambda}{2}$, each pair cancels and the screen is dark.
4. **$\sin\theta = \frac{\lambda}{a}$** for the first minimum.



pair a ray from the top half with one $a/2$ below it
extra path = $(a/2)\sin\theta = \lambda/2$ → dark

5 DIFFRACTION GRATING

SYLLABUS 24.6(B)

RESULT $d\sin\theta = m\lambda$

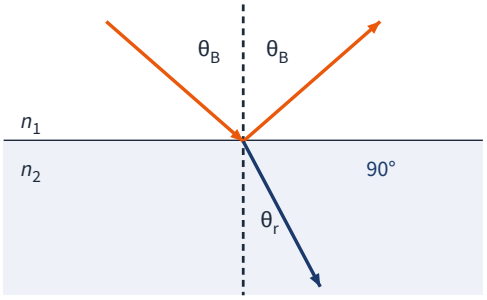
1. Grating spacing $d = \frac{1}{N}$ (N lines per metre).
2. In direction θ, light from adjacent slits has path difference $d\sin\theta$.
3. Bright maxima when all slits are in phase: **$d\sin\theta = m\lambda$** ($m = 0, 1, 2, \ldots$), with m less than $\frac{d}{\lambda}$.

6 BREWSTER'S LAW

SYLLABUS 24.7(F)

RESULT $\tan\theta_B = \frac{n_2}{n_1}$

1. At the Brewster angle the reflected light is completely plane polarised and the reflected and refracted rays are at 90°.
2. $\theta_B + \theta_r = 90°$.
3. Snell's law: $n_1\sin\theta_B = n_2\sin\theta_r = n_2\sin(90° - \theta_B) = n_2\cos\theta_B$.
4. **$\tan\theta_B = \frac{n_2}{n_1}$**.



7 MALUS'S LAW

SYLLABUS 24.7

RESULT $I = I_0 \cos^2\theta$

1. Plane-polarised light of amplitude A_0 (intensity I_0) falls on an analyser whose transmission axis makes angle θ with the plane of polarisation.
2. Resolve the amplitude: only the component parallel to the transmission axis passes, so the amplitude after the analyser is $A = A_0 \cos\theta$.
3. Intensity is directly proportional to the square of the amplitude: $\frac{I}{I_0} = \frac{A^2}{A_0{}^2} = \cos^2\theta$.
4. **$I = I_0 \cos^2\theta$**.

CHAPTER 25 · DERIVATIONS

QUANTUM PHYSICS

6 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 DE BROGLIE WAVELENGTH IN TERMS OF KINETIC ENERGY

SYLLABUS 25.2

RESULT $\lambda = \dfrac{h}{\sqrt{2mK}}$

1. de Broglie relation: $\lambda = \dfrac{h}{p}$, with momentum $p = mv$.
2. Square the momentum: $p^2 = m^2v^2 = m(mv^2)$.
3. Kinetic energy $K = \frac{1}{2}mv^2$, so $mv^2 = 2K$.
4. Substitute mv^2: $p^2 = 2mK$, so $p = \sqrt{2mK}$ and $\boldsymbol{\lambda = \dfrac{h}{\sqrt{2mK}}}$.

2 DE BROGLIE WAVELENGTH OF A PARTICLE ACCELERATED THROUGH A POTENTIAL DIFFERENCE

SYLLABUS 25.2

RESULT $\lambda = \dfrac{h}{\sqrt{2meV}}$

1. de Broglie relation: $\lambda = \dfrac{h}{\sqrt{2mK}}$.
2. A particle of charge e accelerated from rest through a potential difference V: kinetic energy gained = loss of electric potential energy, $K = eV$.
3. $\boldsymbol{\lambda = \dfrac{h}{\sqrt{2meV}}}$.

3 RADII OF BOHR ORBITS

SYLLABUS 25.3(B)

RESULT $r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}$

1. The electric attraction provides the centripetal force: $\dfrac{e^2}{4\pi\varepsilon_0 r^2} = \dfrac{mv^2}{r}$ … (1)
2. Bohr's postulate (quantised angular momentum): $mvr = \dfrac{nh}{2\pi}$, so $v = \dfrac{nh}{2\pi mr}$ … (2)
3. Substitute (2) into (1): $\dfrac{e^2}{4\pi\varepsilon_0 r} = mv^2 = \dfrac{n^2h^2}{4\pi^2mr^2}$.
4. $\boldsymbol{r_n = \dfrac{\varepsilon_0 n^2 h^2}{\pi m e^2}}$; $r_1 = 5.3 \times 10^{-11}$ m.

4 ENERGY LEVELS OF HYDROGEN

SYLLABUS 25.3(C)

RESULT $E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}$

1. From (1): $KE = \frac{1}{2}mv^2 = \dfrac{e^2}{8\pi\varepsilon_0 r}$.
2. $PE = -\dfrac{e^2}{4\pi\varepsilon_0 r}$.
3. Total $E = KE + PE = -\dfrac{e^2}{8\pi\varepsilon_0 r}$.
4. Substitute r_n: $\mathbf{E_n = -\dfrac{me^4}{8\varepsilon_0^2 h^2 n^2}} = -\dfrac{13.6}{n^2}$ eV.

5 MINIMUM X-RAY WAVELENGTH

SYLLABUS 25.4(C)

RESULT $\lambda_{min} = \dfrac{hc}{eV}$

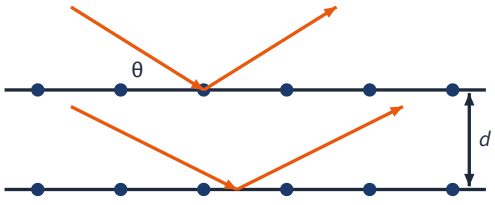
1. Electrons accelerated through p.d. V gain kinetic energy eV.
2. The shortest wavelength is emitted when all of this energy goes into one photon: $eV = hf_{max} = \dfrac{hc}{\lambda_{min}}$.
3. $\mathbf{\lambda_{min} = \dfrac{hc}{eV}}$ (independent of the target metal).

6 BRAGG'S LAW

SYLLABUS 25.4(E)

RESULT $2d \sin\theta = m\lambda$

1. X-rays are reflected by two adjacent atomic planes a distance d apart, at glancing angle θ.
2. The ray from the lower plane travels an extra path $2d \sin\theta$.
3. Constructive interference when the extra path is a whole number of wavelengths: $\mathbf{2d \sin\theta = m\lambda}$.



extra path of lower ray = $2d \sin\theta$

CHAPTER 26 · DERIVATIONS

NUCLEAR PHYSICS

2
derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 RADIOACTIVE DECAY LAW

SYLLABUS 26.2(F)

RESULT $N = N_0 e^{-\lambda t}$

1. Decay law: $\frac{dN}{dt} = -\lambda N$.
2. Separate: $\frac{dN}{N} = -\lambda\, dt$.
3. Integrate from N_0 at $t = 0$: $\ln \frac{N}{N_0} = -\lambda t$.
4. **$N = N_0 e^{-\lambda t}$.**

2 HALF-LIFE

SYLLABUS 26.2(G)

RESULT $t_{½} = \frac{\ln 2}{\lambda}$

1. At $t = t_{½}$, $N = \frac{N_0}{2}$.
2. $\frac{N_0}{2} = N_0 e^{-\lambda t_{½}}$, so $e^{\lambda t_{½}} = 2$.
3. $\lambda t_{½} = \ln 2$, so **$t_{½} = \frac{\ln 2}{\lambda}$** $\approx \frac{0.693}{\lambda}$.